Line integrals

In mechanics line integrals are used to compute work.

In electricity and magnetism, they provide simple methods for computing the electric and magnetic fields for simple geometries.

The line integral most frequently encountered in an oriented one in which the path "C" is directed and the integrand is the dot product between the vector function $\vec{F}(r)$ and the tangent of the path $d\vec{r}$. It is usually written in the form $\int_C \vec{F} \cdot d\vec{r}$

Any integral which is to be evaluated along a curve is called the line integral.

If $\vec{F}(x,y,z)$ or $\vec{F}(r)$ is a vector point function and C is any curve then $\int_C \vec{F} \cdot d\vec{r}$ is called the line integral or tangential line integral.

Note:-

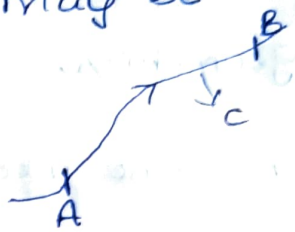
(1) If $\vec{F} = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$, $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$

then, $\int_C \vec{F} \cdot d\vec{r} = \int_C (F_1\hat{i} + F_2\hat{j} + F_3\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k})$

$$= \int_C F_1 dx + F_2 dy + F_3 dz$$

2) Here C is called the path of integration. If the Path of integration is clear, then $\int_C \vec{F} \cdot d\vec{r}$ may be written as $\int_A^B \vec{F} \cdot d\vec{r}$



3) When the path of integration is a closed curve, this fact is indicated by using $\oint \vec{F} \cdot d\vec{r}$

Closed curve is a curve that forms a path whose starting point is also its ending point.

Ex: Circle, triangle, Rectangle, ellipse.

4) If $\vec{F}$ represents the Variable force acting on a Particle along arc AB, then the total work done $= \int_A^B \vec{F} \cdot d\vec{r}$

5) If $\vec{F}$ represents the velocity of fluid, then $\oint_C \vec{F} \cdot d\vec{r}$ is called the circulation.

6) If the circulation of $\vec{F}$ round every closed curve is zero then $\vec{F}$ is said to be irrotational or conservative.

OR

$\vec{F}$ is said to be irrotational $\oint_C \vec{F} \cdot d\vec{r} = 0$

7) The line integral is said to be independent of the path if $\nabla \times \vec{F} = 0$.

① Evaluate $\int \vec{F} \cdot d\vec{r}$, if $\vec{F} = 3xy\hat{i} - y^2\hat{j}$ and C is the curve $y = 2x^2$ in the xy plane from $(0,0)$ to $(1,2)$.

Soln:- the path of integration C lies on the xy-plane

we take $\vec{r} = x\hat{i} + y\hat{j}$

$$d\vec{r} = dx\hat{i} + dy\hat{j}$$

Also we have $y = 2x^2 \rightarrow ① \Rightarrow dy = 4x\,dx \rightarrow ②$

x varies from 0 to 1

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int (3xy\hat{i} - y^2\hat{j}) \cdot (dx\hat{i} + dy\hat{j})$$

$$= \int 3xy\,dx - y^2\,dy$$

$$= \int_0^1 3x(2x^2)\,dx - (4x^4)(4x)\,dx \quad (\text{By } ① \text{ and } ②)$$

$$= \int_0^1 (6x^3 - 16x^5)\,dx$$

$$= \frac{6x^4}{4} - \frac{16x^6}{6}\Big]_0^1$$

$$= \frac{3}{2} - \frac{8}{3}$$

$$= \frac{-7}{6} //$$

(2) If $\vec{F} = y\hat{i} - x\hat{j}$, evaluate $\int_C \vec{F} \cdot d\vec{r}$ from $(0,0)$ to $(1,1)$ along the paths C:

(a) the parabola $y = x^2$

(b) the line from $(0,0)$ to $(1,0)$ and then to $(1,1)$

Soln:- we have $\vec{r} = x\hat{i} + y\hat{j}$

$$d\vec{r} = dx\hat{i} + dy\hat{j}$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int_C (y\hat{i} - x\hat{j}) \cdot (dx\hat{i} + dy\hat{j})$$

$$= \int_C (y\,dx - x\,dy)$$

(a) Here C is the arc of the parabola $y = x^2$.

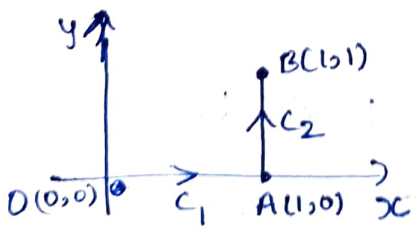
$$y = x^2 \Rightarrow dy = 2x\,dx$$

x varies from 0 to 1.

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int_{x=0}^{x=1} [x^2\,dx - x(2x)\,dx]$$

$$= \int_0^1 -x^2\,dx = -\frac{x^3}{3}\Big|_0^1 = -\frac{1}{3}$$

(b)



let $O(0,0)$, $A = (1,0)$ and $B = (1,1)$

on the line OA, $y = 0 \Rightarrow dy = 0$

and x varies from $x = 0$ to $x = 1$

On the line AB, $x = 1 \Rightarrow dx = 0$

y varies from 0 to 1

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int_{x=0}^{x=1} (0 \cdot dx - x(0)) + \int_{y=0}^{y=1} y(0) - (1)\, dy$$

$$= 0 + \int_0^1 -dy$$

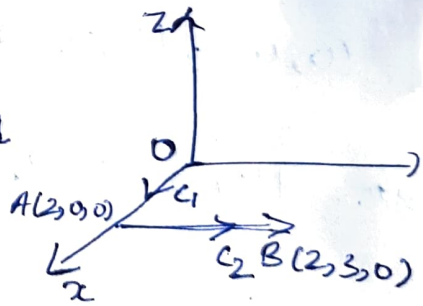
$$= -y\Big|_0^1$$

$$= -1.$$

(3) If $\vec{F} = (2x+y)\hat{i} + (3y-x)\hat{j} + 3z\hat{k}$, evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is composed of the straight line segments OA and AB where O is $(0,0,0)$, A is $(2,0,0)$ and B is $(2,3,0)$

Soln:- $$\int_C \vec{F} \cdot d\vec{r} = \int_C (2x+y)\,dx + (3y-x)\,dy + 3z\,dz$$

Here C is composed of C_1 (i.e OA) and C_2 (i.e AB)



Along C_1; $y = 0, z = 0 \Rightarrow dy = 0, dz = 0$

x varies from 0 to 2

$$\therefore \int_{C_1} \vec{F} \cdot d\vec{r} = \int_0^2 2x\,dx = x^2\Big]_0^2 = 4 \rightarrow (1)$$

Along C_2, $x=2$, $z=0$

$dx=0$, $dz=0$

y varies from 0 to 3

$$\therefore \int_{C_2} \vec{F}\cdot d\vec{r} = \int_0^3 (3y-2)\,dy = \left[\frac{3}{2}y^2 - 2y\right]_0^3$$

$$= \frac{27}{2} - 6 = \frac{15}{2}$$

$$\therefore \int_C \vec{F}\cdot d\vec{r} = \int_{C_1} \vec{F}\cdot d\vec{r} + \int_{C_2} \vec{F}\cdot d\vec{r}$$

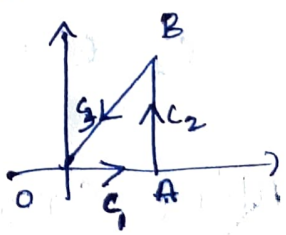
$$= 4 + \frac{15}{2}$$

$$= \frac{23}{2}$$

(4) If $\vec{F} = (2x+y^2)\hat{i} + (3y-4x)\hat{j}$ evaluate $\oint_C \vec{F}\cdot d\vec{r}$ around the triangle in the xy-plane with vertices $(0,0)$, $(2,0)$ and $(2,1)$.

Soln: Here C consists of C_1 (OA), C_2 (AB), C_3 (BO)

$$\therefore \int_C \vec{F}\cdot d\vec{r} = \int_{C_1} \vec{F}\cdot d\vec{r} + \int_{C_2} \vec{F}\cdot d\vec{r} + \int_{C_3} \vec{F}\cdot d\vec{r} \rightarrow (1)$$



Along C_1, $y=0$, $dy=0$, x varies from 0 to 2

$$\int_{C_1} \vec{F}\cdot d\vec{r} = \int_0^2 2x\,dx = x^2\Big]_0^2 = 4 \rightarrow (2)$$

Along C_2, $x=2$, $dx=0$, y varies from 0 to 1

$$\int_{C_2} \vec{F}\cdot d\vec{r} = \int_0^1 (3y-8)\,dy = \left[\frac{3}{2}y^2 - 8y\right]_0^1$$

$$= \frac{3}{2} - 8$$

$$= -13/2 \rightarrow (3)$$

Along C_3, $y = \frac{1}{2}x$, $dy = \frac{1}{2}dx$, x varies from 2 to 0.

$$\int_{C_3} \vec{F}\cdot d\vec{r} = \int_2^0 (2x + \frac{1}{4}x^2)\,dx + (\frac{3}{2}x - 4x)\times\frac{1}{2}dx$$

$$= \int_2^0 \left[\frac{3}{4}x + \frac{1}{4}x^2\right]$$

$$= \frac{3}{8}x^2 + \frac{x^3}{12}\Big]_2^0$$

$$= -\frac{13}{6}$$

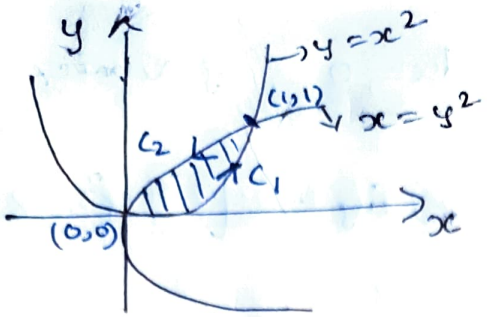
$\therefore$ (1) $\Rightarrow$ $\int_C \vec{F}\cdot d\vec{r} = 4 - \frac{13}{2} - \frac{13}{6}$

$$= -\frac{14}{3}.$$

(5) Evaluate $\int_C \vec{F}\cdot d\vec{r}$ where C is the closed path formed by the arcs $y=x^2$ and $y^2=x$ and

$$\vec{F} = (2x+y^2)\hat{i} + (3y-4x)\hat{j}$$

Soln:-



$y=x^2$; $y^2=x$ $\Rightarrow$ $x^4=x \Rightarrow x(x^3-1)=0$

$x=0$ or 1

$$\int_C \vec{F}\cdot d\vec{r} = \int_{C_1} \vec{F}\cdot d\vec{r} + \int_{C_2} \vec{F}\cdot d\vec{r} \rightarrow (1)$$

Along C_1, $y=x^2 \Rightarrow dy = 2x\,dx$

and x varies from 0 to 1

$$\int_{C_1} \vec{F}\cdot d\vec{r} = \int_{x=0}^{x=1} (2x+x^4)\,dx + (3x^2-4x)\,2x\,dx$$

$$= \int_{x=0}^{x=1} (2x+x^4+6x^3-8x^2)\,dx$$

$$= x^2 + \frac{x^5}{5} + \frac{6x^4}{4} - \frac{8x^3}{3}\Big]_0^1$$

$$= \frac{1}{30} \rightarrow (2)$$

Along C_2, $y^2=x \Rightarrow dx = 2y\,dy$ and y varies from 1 to 0.

$$\therefore \int_{C_2} \vec{F}\cdot d\vec{r} = \int_{y=1}^{y=0} (2y^2+y^2)\,2y\,dy + (3y-4y^2)\,dy$$

$$= \int_{y=1}^{0} (6y^3+3y-4y^2)\,dy = -\frac{5}{3} \rightarrow (3)$$

$$\therefore (1) \Rightarrow \int_C \vec{F}\cdot d\vec{r} = \frac{1}{30} - \frac{5}{3} = -\frac{49}{30}$$

(6) If $\vec{F} = (3x^2+6y)\hat{i} - 14yz\hat{j} + 20xz^2\hat{k}$ evaluate $\int_C \vec{F}\cdot d\vec{r}$ from $(0,0,0)$ to $(1,1,1)$ along the $x=t$, $y=t^2$, $z=t^3$.

Soln:- Along C, $x=t$, $y=t^2$, $z=t^3$

$dx=dt$, $dy=2t\,dt$, $dz=3t^2\,dt$

t varies from 0 to 1

$$\int_C \vec{F}\cdot d\vec{r} = \int_C (3x^2+6y)\,dx - 14yz\,dy + 20xz^2\,dz$$

$$= \int_{t=0}^{t=1} (3t^2+6t^2)\,dt - 14(t^2)(t^3)2t\,dt + 20t(t^6)3t^2\,dt$$

$$= \int_{t=0}^{t=1} (9t^2 - 28t^6 + 60t^9)\,dt$$

$$= 3t^3 - 4t^7 + 6t^{10}\Big]_0^1$$

$$= 3 - 4 + 6 = 5 //$$

(7) Evaluate $\int_{(1,1,1)}^{(4,0,-2)} \left[\frac{x-1}{2}\hat{i} + (1-y)^2\hat{j} + \left(\frac{z-1}{3}\right)\hat{k}\right]\cdot d\vec{r}$

along the Straight path.

Soln:- The Straight line joining $(1,1,1)$ to $(4,0,-2)$

is $$\frac{x-1}{4-1} = \frac{y-1}{0-1} = \frac{z-1}{-2-1} = t$$

$$\frac{x-1}{3} = \frac{y-1}{-1} = \frac{z-1}{-3} = t$$

$\Rightarrow x-1=3t, \quad y-1=-t, \quad z-1=-3t$

$dx=3dt, \quad dy=-dt \quad dz=-3dt$

x varies from 1 to 4.

$\Rightarrow t$ varies from 0 to 1 ($\because x-1=3t$ when $x=1, t=0$; $x=4, t=1$)

$$\therefore \int_C \left[\left(\frac{x-1}{2}\right)\hat{i}+(1-y)^2\hat{j}+\left(\frac{z-1}{3}\right)\hat{k}\right]\cdot d\vec{r}$$

$$= \int_C \left[\frac{3t}{2}\hat{i}+(t)^2\hat{j}-\frac{3t}{3}\hat{k}\right]\cdot(dx\hat{i}+dy\hat{j}+dz\hat{k})$$

$$= \int_{t=0}^{t=1} \left[\frac{3t}{2}\hat{i}+(t^2)\hat{j}-\frac{3t}{3}\hat{k}\right]\cdot(3dt\hat{i}-dt\hat{j}-3dt\hat{k})$$

$$= \int_{t=0}^{t=1} \left[\frac{9t}{2}-t^2+3t\right]dt$$

$$= \frac{9t^2}{4}-\frac{t^3}{3}+\frac{3t^2}{2}\Big]_0^1$$

$$= \frac{9}{4}-\frac{1}{3}+\frac{3}{2}$$

$$= \frac{15}{4}-\frac{1}{3}$$

$$= \frac{45-4}{12} = \frac{41}{12}$$

$$= \frac{41}{12}.$$

(8) Compute $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \frac{y\hat{i} - x\hat{j}}{x^2+y^2}$ and C is the circle $x^2+y^2=1$ traversed counter clockwise.

Soln:- Let us consider the parametric Eqn of circle

$$x = \cos\theta \qquad y = \sin\theta$$

$$dx = -\sin\theta\, d\theta, \qquad dy = \cos\theta\, d\theta$$

$$\theta :- \ 0 \text{ to } 2\pi$$

$$\therefore \oint_C \vec{F} \cdot d\vec{r} = \int_C \left(\frac{y\hat{i} - x\hat{j}}{x^2+y^2}\right) \cdot (dx\hat{i} + dy\hat{j})$$

$$= \int_C y\,dx - x\,dy$$

$$= \int_{\theta=0}^{\theta=2\pi} \sin\theta(-\sin\theta)\,d\theta - \cos\theta(\cos\theta)\,d\theta$$

$$= \int_0^{2\pi} (-\sin^2\theta - \cos^2\theta)\,d\theta$$

$$= -\int_0^{2\pi} (\sin^2\theta + \cos^2\theta)\,d\theta$$

$$= -\int_0^{2\pi} d\theta$$

$$= -[\theta]_0^{2\pi}$$

$$= -2\pi$$

(9) Determine whether the line integral

$\int (6xy+z^3)\,dx + (3x^2-z)\,dy + (3xz^2-y)\,dz$ is

independent of path of integration

Soln:- To prove the path of integration is independent, it is sufficient if we prove $\nabla \times \vec{F} = 0$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 6xy+z^3 & 3x^2-z & 3xz^2-y \end{vmatrix}$$

$$= (-1+1)\hat{i} - (3z^2-3z^2)\hat{j} + (6x-6x)\hat{k}$$

$$= 0.$$

Hence the line integral is independent of path of integration.

(10) Find the work done in moving a particle in the force field $\vec{F} = 3x^2\hat{i} + (2xz - y)\hat{j} + z\hat{k}$ along the curve defined by $x^2 = 4y$, $3x^3 = 8z$ from $x=0$ to $x=2$.

Soln:. Let C be the curve, $y = \frac{x^2}{4}$, $3x^3 = 8z$

$\Rightarrow dy = \frac{x}{2}dx$ $\quad \Rightarrow z = \frac{3x^3}{8}$

$\Rightarrow dz = \frac{9x^2}{8}dx$

x varies from 0 to 2.

Work done $= \int_C \vec{F} \cdot d\vec{r}$

$$= \int_C 3x^2 dx + (2xz - y)dy + z\,dz$$

$$= \int_{x=0}^{x=2} 3x^2 dx + \left[2x\left(\frac{3x^3}{8}\right) - \frac{x^2}{4}\right]\frac{x}{2}dx + \frac{3x^3}{8} \times \frac{9x^2}{8}dx$$

$$= \int_0^2 \left(3x^2 + \frac{3x^5}{8} - \frac{x^3}{8} + \frac{27}{64}x^5\right)dx$$

$$= \int_0^2 \left(3x^2 - \frac{x^3}{8} + \frac{51}{64}x^5\right)dx$$

$$= x^3 - \frac{x^4}{32} + \frac{51}{64}\frac{x^6}{6}\Big|_0^2$$

$$= 8 - \frac{1}{2} + \frac{17}{2} = 16$$